

TDMA Schedule Planner and Optimizer

Distance-2 Graph Colouring with EMANE Integration

Project Report

Project	TDMA Schedule Planner and Optimizer
Primary language	Python
Network size	16 static radio nodes
Communication range	500 m
Scheduling model	Distance-2 graph colouring
EMANE model	TDMA Radio Model
Final schedule	5 slots
Validation	VALID — 0 violations

Prepared for project submission

1. Abstract

This project implements a centralised TDMA Schedule Planner and Optimizer for a network of 16 static radio nodes. The system accepts node coordinates, constructs a communication graph using a 500 m radio range, derives a conflict graph containing both one-hop and two-hop interference relationships, and assigns TDMA slots through distance-2 graph colouring. The generated schedule is independently validated to ensure that conflicting nodes never share a slot. For the supplied 16-node input, the implementation produces a valid 5-slot schedule with 55 conflict edges and zero schedule violations. The 5-slot result equals the largest-clique lower bound for this input, proving minimality for this particular topology.

The project also contains an EMANE integration layer that maps the computed schedule to an EMANE TDMA schedule XML configuration and includes a 16-NEM test configuration. The EMANE environment accepted a full schedule on NEM 1, and the generated virtual interfaces were available for connectivity checks.

Keywords: TDMA, graph colouring, distance-2 colouring, conflict graph, spatial reuse, EMANE, Python, NetworkX.

2. Problem Statement

In a wireless network, multiple radios cannot transmit simultaneously when their transmissions could interfere with each other. A TDMA system avoids such collisions by dividing time into slots and assigning transmission opportunities to nodes. The challenge is to reuse slots where safe while preventing direct and hidden-terminal interference.

For this project, two nodes are connected in the communication graph when their Euclidean distance is less than or equal to 500 m. The conflict model considers both one-hop neighbours and two-hop neighbours. A two-hop relationship is important because a chain A-B-C can create a hidden-terminal situation: A and C may not hear each other directly, but simultaneous transmissions can interfere at B.

The scheduling problem is therefore represented as graph colouring: every node receives a colour, and the colour represents its TDMA slot. Conflicting nodes must have different colours. Nodes that are sufficiently separated in the conflict graph can reuse the same slot.

3. Objectives

- Read and validate the 16-node coordinate input.
- Construct the 500 m communication graph.
- Construct a one-hop and two-hop conflict graph.
- Generate a TDMA schedule using distance-2 graph colouring.
- Compare multiple colouring heuristics and retain the best valid schedule.
- Calculate a lower bound using the largest clique in the conflict graph.
- Independently validate the final schedule.
- Generate result files, matrices and visualisations.
- Map the schedule to an EMANE TDMA schedule XML configuration.
- Provide automated tests covering the core implementation.

4. System Architecture

The processing pipeline is organised as follows:

```
nodes.json → input_parser → distance → topology → conflict_graph → coloring → scheduler →  
validator/report → results
```

The EMANE integration is connected through the schedule bridge:

```
scheduler → emane_bridge → emane/schedule.xml → EMANE TDMA Radio Model
```

The main implementation is modular. Separate modules handle input parsing, distance calculation, topology construction, conflict generation, colouring, scheduling, validation, reporting, result export and EMANE integration.

5. Methodology

5.1 Input and Distance Calculation

For two nodes with coordinates (x1, y1) and (x2, y2), Euclidean distance is calculated as:

$$d = \sqrt{(x2 - x1)^2 + (y2 - y1)^2}$$

The communication graph uses a boundary-inclusive 500 m threshold. Each node is represented as a graph vertex and an edge is added when the distance is at most the configured range.

5.2 Communication Graph

The supplied topology contains 16 nodes and 23 communication edges. The graph is connected, with node degree ranging from 2 to 4 and an average degree of 2.88.

5.3 Conflict Graph

The conflict graph contains 55 edges: 23 one-hop conflicts and 32 two-hop conflicts. Two nodes connected by a conflict edge are not permitted to share a TDMA slot.

5.4 Distance-2 Colouring

The implementation evaluates several ordering/colouring strategies including greedy input order, largest-first, smallest-last, DSATUR and random restarts. The best schedule is then subjected to slot compaction and independent validation.

5.5 Lower Bound and Optimality for the Supplied Input

A clique in the conflict graph is a set of mutually conflicting nodes and therefore requires at least one distinct slot per node. The implementation calculates the largest clique as a lower bound. For the supplied input, the lower bound is 5 and the schedule also uses 5 slots. Therefore the result is proven minimal for this input.

6. Input Topology and Results

Metric	Result
Nodes	16
Radio range	500 m
Communication edges	23
1-hop conflicts	23
2-hop conflicts	32
Total conflict edges	55
Slots used	5
Lower bound	5
Validation violations	0
Final status	VALID

6.1 Node-to-Slot Mapping

Slot	Nodes
0	Node_01, Node_04, Node_10, Node_16
1	Node_02, Node_08, Node_09, Node_15
2	Node_03, Node_05, Node_11
3	Node_06, Node_12, Node_13
4	Node_07, Node_14

All five slots are reused. The closest pair of nodes sharing a slot is three hops apart, which is outside the one-hop/two-hop conflict definition.

6.2 Slot × Node Matrix

Slot \ Node	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16
00	1	0	0	1	0	0	0	0	0	1	0	0	0	0	0	1
01	0	1	0	0	0	0	0	1	1	0	0	0	0	0	1	0
02	0	0	1	0	1	0	0	0	0	0	1	0	0	0	0	0
03	0	0	0	0	0	1	0	0	0	0	0	1	1	0	0	0
04	0	0	0	0	0	0	1	0	0	0	0	0	0	1	0	0

7. Network Topology

Generated result image: **network_topology.png**. The image is included in the project repository under results/.

8. Conflict Graph

Generated result image: **conflict_graph.png**. The image is included in the project repository under results/.

9. TDMA Schedule Visualisation

Generated result image: **tdma_schedule.png**. The image is included in the project repository under results/.

10. Spatial Reuse Visualisation

Generated result image: **spatial_reuse.png**. The image is included in the project repository under results/.

11. Software Testing

The project includes automated tests covering the core scheduling and validation pipeline. The final recorded test run reported:

47 passed, 7 subtests passed in 1.52s

Area	Coverage
Topology	Range boundary and communication-edge construction
Conflict detection	One-hop and two-hop relationships
Colouring	Spatial reuse and schedule generation
Input parsing	Invalid JSON and invalid coordinates
Validation	Valid and invalid schedules
Result export	Schedule and statistics outputs
CLI	Command-line execution and exit behaviour
EMANE XML	Schedule/configuration related checks

The test result is retained in docs/proof/test_output.txt in the project submission.

12. EMANE Integration

The project contains an EMANE integration layer for translating the calculated TDMA schedule into an EMANE TDMA schedule XML configuration. The configuration includes 16 NEM definitions, virtual transport interfaces, platform configuration and the TDMA radio model.

12.1 EMANE Schedule Parameters

Parameter	Configured value
Frames	1
Slots	5
Slot duration	1500 μ s
Slot overhead	50 μ s
Bandwidth	1 MHz
Frequency	2.4 GHz

12.2 Scheduler Acceptance Proof

The recorded EMANE scheduler status for NEM 1 shows that the full schedule was accepted. No frame-index, slot-index, update-order or other schedule rejection was recorded.

EMANE statistic	Value
scheduleAcceptFull	1
scheduleAcceptUpdate	0
scheduleRejectFrameIndexRange	0
scheduleRejectOther	0

EMANE statistic	Value
scheduleRejectSlotIndexRange	0
scheduleRejectUpdateBeforeFull	0

12.3 EMANE Schedule Table

Index	Frame	Slot	Type	Frequency	Data Rate
0	0	0	TX	2.4 GHz	1 MHz
1	0	1	RX	2.4 GHz	—
2	0	2	RX	2.4 GHz	—
3	0	3	RX	2.4 GHz	—
4	0	4	RX	2.4 GHz	—

The project also recorded successful ICMP connectivity checks from the Node 1 interface to the interfaces corresponding to Nodes 2, 3 and 16, each with 5 packets transmitted and 0% packet loss.

13. Reproducibility and Evidence

The submission contains the following evidence files under docs/proof/:

- algorithm_output.txt — complete algorithm and validation output.
- test_output.txt — automated test result.
- emane_interfaces.txt — EMANE virtual interfaces.
- emane_scheduler_status.txt — scheduler acceptance/rejection counters.
- emane_schedule_table.txt — EMANE ScheduleInfoTable and StructureInfoTable.
- ping_node1_to_node2.txt — connectivity test.
- ping_node1_to_node3.txt — connectivity test.
- ping_node1_to_node16.txt — connectivity test.

The results directory also contains schedule.csv, schedule.json, statistics.json, final_report.txt and the generated visualisations.

14. Limitations

- The scheduler uses a simplified static topology based on fixed node coordinates.
- The conflict model considers one-hop and two-hop interference rather than a full physical SINR model.
- The schedule is node-based rather than traffic-demand or link-demand weighted.
- No fairness or traffic prioritisation mechanism is implemented.
- The general colouring heuristic is not guaranteed to find a globally minimum colouring for arbitrary graphs.
- The 5-slot minimality proof applies specifically to the supplied input because its lower bound equals the generated slot count.

15. Future Enhancements

- Traffic-aware and link-level TDMA scheduling.
- SINR-based interference modelling.

- Dynamic rescheduling when nodes move.
- Fairness and priority-aware slot assignment.
- Exact ILP/SAT optimisation for small networks.
- Local-search or tabu-search optimisation for larger networks.
- Extended EMANE packet-level experiments under controlled traffic loads.

16. Conclusion

The completed implementation provides an end-to-end TDMA scheduling workflow for the supplied 16-node topology. It constructs the communication and conflict graphs, performs distance-2 colouring, generates a reusable TDMA schedule, independently validates the result, exports machine-readable results and provides EMANE configuration support.

For the supplied input, the final schedule uses 5 slots, has a lower bound of 5 and records zero validation violations. The automated test suite reports 47 passing tests and 7 passing subtests. The EMANE evidence records acceptance of the full schedule on NEM 1 and successful virtual-interface connectivity checks.

17. Project Structure

```
TDMA Project/
■■■ main.py
■■■ src/ # scheduling implementation
■■■ tests/ # automated tests
■■■ input/ # node coordinate inputs
■■■ emane/ # EMANE configuration and schedule
■■■ results/ # generated schedules, reports and graphs
■■■ docs/ # documentation and proof outputs
■■■ visualization/ # graph visualisation scripts
■■■ requirements.txt
■■■ run_all.py
■■■ run_complete.sh
```

18. Submission Reference

GitHub repository: <https://github.com/Manjari-E/TDMA-Scheduling-and-EMANE>